

INDUSTRIAL INTERNSHIP REPORT

On

GROCERY DELIVERY APPLICATION

Prepared By

Praveen Kumar

Domain: Full Stack Development

Date of Submission: 29 Sept. 2026

EXECUTIVE SUMMARY

This report presents the work completed during the Industrial Internship conducted through **upskill Campus and The IoT Academy**, in collaboration with **UniConverge Technologies Pvt. Ltd. (UCT)**.

The internship provided an opportunity to work on a practical Full Stack Development project based on the problem statement of developing a **Grocery Delivery Application**.

The proposed application is designed to provide an online platform through which users can browse grocery products, view product information and prices, add products to a shopping cart, place orders, select suitable delivery slots, and proceed through the payment and order-confirmation workflow.

The project involved understanding the requirements, planning the application architecture, designing the database, developing different application modules, integrating these modules, and performing testing and debugging.

The internship provided practical exposure to software development concepts such as database management, CRUD operations, module integration, order management, application testing, and problem-solving. It also helped in understanding how theoretical concepts can be applied to develop a practical industry-oriented application.

TABLE OF CONTENTS

- 1. Preface**
 - 2. Introduction**
 - 2.1 About UniConverge Technologies Pvt. Ltd.
 - 2.2 About upskill Campus
 - 2.3 About The IoT Academy
 - 2.4 Objectives of the Internship
 - 2.5 References
 - 2.6 Glossary
 - 3. Problem Statement**
 - 4. Existing and Proposed Solution**
 - 4.1 Code Submission
 - 4.2 Report Submission
 - 5. Proposed Design / Model**
 - 5.1 High-Level Design
 - 5.2 Low-Level Design
 - 5.3 Interfaces
 - 6. Performance Test**
 - 6.1 Test Plan / Test Cases
 - 6.2 Test Procedure
 - 6.3 Performance Outcome
 - 7. My Learnings**
 - 8. Future Work Scope**
-

1. PREFACE

Industrial internships provide students with an opportunity to gain practical exposure to technologies and development practices used in real-world projects. The internship helps bridge the gap between theoretical knowledge acquired in academic courses and its practical application.

During this internship, I worked on a **Grocery Delivery Application** under the Full Stack Development domain. The project provided practical exposure to application development, database management, module integration, testing, and debugging.

The project was carried out through **upskill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt. Ltd. (UCT)**. The sample internship material describes the program as an opportunity to work on industrial problem statements and develop practical solutions.

The internship helped me understand the complete development process, beginning with requirement analysis and project planning and continuing through implementation, integration, testing, and documentation.

I would like to express my gratitude to the mentors, trainers, coordinators, and everyone who provided guidance and support throughout the internship.

2. INTRODUCTION

2.1 About UniConverge Technologies Pvt. Ltd. (UCT)

UniConverge Technologies Pvt. Ltd. (UCT) was established in 2013 and works in the **Digital Transformation** domain, with a focus on sustainability and Return on Investment (RoI).

According to the provided internship material, UCT works with technologies including:

- Internet of Things (IoT)
- Cyber Security
- Cloud Computing
- Machine Learning
- Communication Technologies
- Java Full Stack
- Python
- Front-End Technologies

UCT's IoT platform, **UCT Insight**, is described as an IoT platform designed for quick deployment of IoT applications and business/process insights. It uses Java for the backend and ReactJS for the frontend and supports MySQL and various NoSQL databases. The organization also works in areas such as smart factories, LoRaWAN-based solutions, industrial monitoring, smart cities, metering solutions, and predictive maintenance.

2.2 About upskill Campus

upskill Campus (USC), along with The IoT Academy and in association with UniConverge Technologies, facilitated the internship program. The provided material describes USC as a career-development platform focused on personalized executive coaching and scalable career development.

2.3 About The IoT Academy

The IoT Academy is described in the provided material as the EdTech division of UCT. It conducts long-term executive certification programs in collaboration with EICT Academy and IIT institutions in multiple domains.

2.4 Objectives of the Internship

The major objectives of the internship were:

- To gain practical experience of working in an industry-oriented environment.
- To understand and solve real-world problems.
- To improve job prospects through practical technical exposure.
- To develop a better understanding of the selected technical field and its applications.
- To improve communication and problem-solving abilities.

These objectives are also explicitly listed in the provided internship report template.

2.5 References

The following types of resources were used during the project:

1. Internship project requirements provided by the internship program.
 2. Full Stack Development learning resources.
 3. Database and MySQL documentation.
 4. Programming and application-development tutorials.
 5. Technical documentation related to application integration and testing.
-

2.6 Glossary

Term	Meaning
------	---------

Full Stack Development	involving frontend, backend, and database components
------------------------	--

CRUD	Create, Read, Update and Delete
------	---------------------------------

DBMS	Database Management System
------	----------------------------

SQL	Structured Query Language
-----	---------------------------

API	Application Programming Interface
-----	-----------------------------------

Term	Meaning
UI	User Interface
Cart	Temporary collection of products selected for purchase
Order	Record containing products selected by a customer for purchase
Database	Structured collection of application data
Testing	Process of checking whether the application works as expected

3. PROBLEM STATEMENT

The objective of the project is to develop a **Grocery Delivery Application** that allows users to purchase grocery products through an online platform.

A traditional grocery purchasing process requires customers to physically visit a store, identify required products, check their prices and availability, and complete the purchase manually.

The proposed application aims to provide a convenient digital platform where grocery products can be listed along with their quantities and prices. Users should be able to register, browse available products, purchase groceries, select a preferred delivery slot, and proceed to payment.

The project requirements provided in the internship material specifically identify grocery listing, quantity and price information, user registration, purchasing, delivery-slot selection, and payment as important functionality.

Major Requirements

1. User registration and login.
2. Grocery product listing.
3. Product quantity and price information.
4. Product selection.
5. Shopping cart.
6. Order placement.
7. Delivery-slot selection.
8. Payment workflow.

9. Order confirmation.
10. Database storage and management.

4. EXISTING AND PROPOSED SOLUTION

4.1 Existing Solution

Traditional grocery shopping requires customers to physically visit a grocery store and manually select products.

Some existing online grocery platforms provide online purchasing facilities, but the project focuses on understanding and implementing the fundamental workflow of an online grocery-delivery system as an industry-oriented Full Stack Development project.

Limitations of the Traditional Approach

- Requires physical presence at the store.
- Product availability must be checked manually.
- Customers spend additional time travelling and shopping.
- No centralized digital shopping workflow.
- Delivery scheduling is not available in a traditional store environment.

4.2 Proposed Solution

The proposed **Grocery Delivery Application** provides a digital platform for grocery purchasing.

The application allows users to:

- Register and log in.
- Browse grocery products.
- View product information and prices.
- Add products to a cart.
- Modify product quantities.
- View the total amount.
- Place an order.

- Select a delivery slot.
- Proceed through the payment workflow.
- Receive order confirmation.

Value Addition

The project brings the complete grocery-shopping workflow into a single application.

The major value addition is the integration of:

Product Management + Shopping Cart + Order Management + Delivery Scheduling + Payment Workflow

into one application.

4.3 Code Submission

GitHub Repository: [upskillcampus/GroceryDeliveryApplication.py at main · praveen26102005/upskillcampus](https://github.com/praveen26102005/GroceryDeliveryApplication.py)

4.4 Report Submission

Project Report Repository/Link: [upskillcampus/GroceryDeliveryApplication_Praveen_USC_UCT.pdf at main · praveen26102005/upskillcampus](https://github.com/praveen26102005/GroceryDeliveryApplication_Praveen_USC_UCT.pdf)

5. PROPOSED DESIGN / MODEL

The application is designed using a modular approach in which different functionalities are separated into individual components.

5.1 High-Level Design

The high-level workflow of the system can be represented as:

User

↓

Registration / Login

↓

Grocery Product Listing

↓

Product Selection

↓

Shopping Cart

↓

Delivery Slot Selection

↓

Order Placement

↓

Payment

↓

Order Confirmation

↓

Database

The provided report format recommends including a high-level diagram showing the overall system design.

Suggested Diagram

You can create a diagram with these blocks:



Product Selection

|

v

Shopping Cart

|

v

Delivery Slot

Selection

|

v

Order Placement

|

v

Payment

|

v

Order Confirmation

|

v

DATABASE

5.2 Low-Level Design

The application can be divided into the following modules:

1. User Module

Responsible for:

- User registration.
- Login.
- User information.
- Authentication-related operations.

2. Product Module

Responsible for:

- Product information.
- Product names.
- Prices.
- Available quantities.
- Product retrieval and management.

3. Cart Module

Responsible for:

- Adding products.
- Removing products.
- Updating quantities.
- Calculating total price.

4. Order Module

Responsible for:

- Creating orders.
- Storing order details.

- Associating products with orders.
- Maintaining order information.

5. Delivery Module

Responsible for:

- Displaying available delivery slots.
- Selecting a preferred slot.
- Associating the selected slot with an order.

6. Payment Module

Responsible for:

- Maintaining the payment workflow.
- Processing the payment stage.
- Supporting order confirmation after the payment process.

7. Database

The database stores information related to:

- Users
- Grocery products
- Orders
- Order items
- Delivery slots
- Payment/order status

5.3 Interfaces

The application can contain the following primary interfaces:

Login / Registration Interface

Allows new users to register and existing users to log in.

Product Listing Interface

Displays available grocery products with their names, prices, and quantities.

Shopping Cart Interface

Displays selected products, quantities, and total amount.

Delivery Slot Interface

Allows users to select their preferred delivery slot.

Order and Payment Interface

Displays order information and provides the payment workflow.

Order Confirmation Interface

Displays confirmation after successful order processing.

6. PERFORMANCE TEST

The provided internship template emphasizes testing because it demonstrates that the project is being treated as an industry-oriented application rather than only an academic project. It recommends identifying constraints, explaining how they are handled, and documenting test results.

For this project, functional testing was primarily performed.

6.1 Test Plan / Test Cases

Test Case	Expected Result	Status
User registration	User information should be stored successfully	Passed
User login	Valid user should be able to access the application	Passed
Product display	Available products should be displayed	Passed
Add product to cart	Selected product should appear in cart	Passed
Update quantity	Cart quantity and total should update	Passed
Remove product	Product should be removed from cart	Passed
Order creation	Order should be created from cart information	Passed

Test Case	Expected Result	Status
Delivery slot selection	Selected slot should be associated with order	Passed
Payment workflow	Payment stage should be accessible	Passed
Order confirmation	Order confirmation should be generated after completion	Passed

6.2 Test Procedure

Testing was performed module by module and then through the complete application workflow.

Step 1

Tested user registration and login.

Step 2

Verified grocery product listing and product information.

Step 3

Added different products to the shopping cart.

Step 4

Modified quantities and checked total-price calculations.

Step 5

Removed products and verified cart updates.

Step 6

Created orders using selected cart items.

Step 7

Selected delivery slots.

Step 8

Tested the payment workflow.

Step 9

Verified order confirmation and database information.

Step 10

Performed end-to-end testing and corrected identified issues.

6.3 Performance Outcome

The application was tested using different combinations of users, products, quantities, and order scenarios.

The testing helped identify integration and data-handling issues, particularly around cart and order information. These issues were addressed through debugging, database-query checking, and repeated testing.

The final testing process helped verify the overall application workflow:

Registration → Product Browsing → Cart → Delivery Slot → Order → Payment → Confirmation

7. MY LEARNINGS

The internship provided practical exposure to the complete software-development process.

Technical Learning

During the project, I learned and improved my understanding of:

- Full Stack Development.
- Database design.
- MySQL.
- CRUD operations.
- User and product management.
- Shopping-cart implementation.
- Order management.
- Database integration.
- Module integration.
- Application testing.
- Debugging and error handling.

Problem-Solving Learning

The project helped me understand how a large application can be divided into smaller modules.

I learned that individual modules should first be tested separately before being integrated into a complete system.

I also gained experience in identifying problems by tracing data flow between different components and testing the relevant functionality.

Career Development

The internship provided practical experience beyond classroom-based programming. It helped me understand how requirements are converted into application modules and how those modules are integrated into a complete working system.

The internship objectives themselves emphasize practical industry experience, real-world problem solving, improved understanding of applications, and personal development.

Overall, the project improved my confidence in working with full-stack application concepts and database-driven systems.

8. FUTURE WORK SCOPE

The current application can be extended with additional features in the future.

Possible improvements include:

1. Online Payment Gateway

A production-ready payment gateway can be integrated to support secure online payments.

2. Product Search and Filtering

Users can be provided with advanced search and filtering options to find grocery products quickly.

3. Product Categories

Products can be organized into categories such as:

- Fruits and Vegetables
- Dairy Products
- Beverages
- Snacks

- Household Products
- Personal Care

4. Order Tracking

Users could track the current status of their orders.

5. Notifications

Email or push notifications could be introduced for:

- Order confirmation
- Payment status
- Order preparation
- Delivery updates

6. Recommendation System

A recommendation system could suggest products based on previous purchases and user preferences.

7. Admin Dashboard

An administrative dashboard could be developed for:

- Product management
- Inventory management
- Order management
- User management
- Sales analytics

8. Scalability and Security

For production deployment, the application could be improved with stronger authentication, security mechanisms, scalability improvements, optimized database queries, and cloud deployment.

The sample report specifically suggests using the Future Work section for ideas that could not be completed because of time limitations.

CONCLUSION

The **Grocery Delivery Application** provided practical experience in Full Stack Development and demonstrated how different software components can be integrated to solve a practical problem.

The project covered the complete development cycle, including requirement analysis, database planning, application development, module integration, testing, debugging, and documentation.

The internship was valuable in improving my technical knowledge, problem-solving ability, database understanding, and practical development skills. It also provided an understanding of how software applications can be designed and developed in an industry-oriented environment.